

scaleLLMflow

General LLM Workflows for Scientific Quality Scales

Proyecto MQS

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URL <https://github.com/ppmena/scaleLLMflow>

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1 Package scaleLLMflow

1.1 Description

Reusable helpers for registering scientific quality scales, resolving prompts by model, sending article text to Gemini or OpenAI, validating strict JSON responses, parsing item scores, comparing them with reviewed references, and writing audit and evidence files.

The package deliberately separates three layers: the formal scale definition in metadata, the operational prompt sent to the model, and the numeric coding used for analysis. The bundled MQS registry is the first example.

1.2 Two workflow modes

Free mode The model response is scored and preserved for inspection without assuming that a previous answer is correct.

Reference mode A reviewed score vector is supplied and every returned item is compared with it. This mode reports agreement and mismatches while preserving the model's evidence and reasoning.

1.3 Credentials and provider configuration

Gemini credentials are read from `GEMINI_API_KEY` or `GOOGLE_GEMINI_KEY`. OpenAI credentials are read from `OPENAI_API_KEY`; `OPENAI_PROJECT_ID` may also be supplied. A project-level `.Renviron` file is convenient for interactive work in RStudio and should not be committed.

2 Registry and scale helpers

`available_scales(registry_dir = NULL)`

2.1 Description

Lists scales available in the prompt registry.

2.2 Arguments

registry_dir Optional registry root. The bundled registry is used when omitted.

2.3 Value

A character vector of scale identifiers.

`available_models(scale, registry_dir = NULL)`

2.4 Description

Lists model prompt folders registered for a scale.

2.5 Arguments

scale Scale identifier, such as `mqs`.

registry_dir Optional registry root.

2.6 Value

A character vector of model identifiers.

```
resolve_prompt(scale, model, registry_dir = NULL)
```

2.7 Description

Resolves the most appropriate registered prompt and its metadata. Matching proceeds from the exact model to family, numeric-version, generic, and nearest available fallbacks.

2.8 Arguments

scale Scale identifier.

model Requested model identifier.

registry_dir Optional registry root.

2.9 Value

A resolved registry record containing prompt text, metadata, scale definition, response schema, and provenance fields.

```
audit_model_registry(registry_dir = NULL, output_dir = NULL, apply_unifications = FALSE)
```

2.10 Description

Checks every scale and model registry entry for required files, valid metadata, a complete item definition, a coherent response schema, and prompt hashes. Exact duplicate prompts are grouped and a unification plan is reported.

2.11 Arguments

registry_dir Optional registry root.

output_dir Optional directory for audit CSV reports.

apply_unifications If TRUE, moves redundant prompt folders to a timestamped backup after reporting the plan.

2.12 Value

An audit result with issue and success counts, duplicate groups, and proposed or applied unifications.

3 Prompt and text preparation

```
build_scale_prompt(prompt_text, article_text)
```

3.1 Description

Combines an operational scale prompt with the article text.

3.2 Arguments

prompt_text Registered prompt text.

article_text Article text to evaluate.

3.3 Value

The complete provider prompt as a character string.

```
extract_article_text(file_path, filetype = "auto", strip_references = TRUE)
```

3.4 Description

Reads local Markdown or text articles and optionally removes the references section. PDF input is supported through **extract_pdf_text**.

3.5 Arguments

file_path Local article path.

filetype One of `auto`, `txt`, `md`, or `pdf`.

strip_references Whether to remove references before model submission.

3.6 Value

Article text.

```
extract_pdf_text(pdf_path, strip_references = TRUE)
```

3.7 Description

Extracts text from a local PDF using `pdftools`.

3.8 Arguments

pdf_path Local PDF path.

strip_references Whether to remove references.

3.9 Value

Article text.

```
strip_references_section(article_text)
```

3.10 Description

Removes a detected references or bibliography section.

3.11 Value

Text with the trailing references section removed when detected.

4 Parsing and provider calls

`parse_scale_scores(text, items = 1:10, metadata = NULL)`

4.1 Description

Validates and parses a strict JSON scale response. Fenced JSON is normalized in a controlled way before validation. The registered metadata defines required items, allowed values, excluded items, and total-score rules.

4.2 Arguments

text Raw model response.

items Expected item identifiers.

metadata Optional resolved metadata and JSON Schema.

4.3 Value

A structured result with item scores, total, validation status, evidence, reasons, and the raw response.

```
run_llm(prompt, provider = "gemini", model = "gemini-3.6-flash", temperature = 0, top_p = 0.1, timeout = 300, api_key = NULL, project_id = NULL, max_retries = 3, retry_wait_seconds = 1, retry_backoff = 2, rate_limit_seconds = 0)
```

4.4 Description

Calls Gemini through the Interactions API or OpenAI through the Responses API. It applies rate limiting, retries transient failures with exponential backoff, records detailed errors, and returns the raw response needed for auditability.

4.5 Arguments

prompt Prompt sent to the provider.

provider gemini, openai, or chatgpt.

model Provider model identifier.

temperature Sampling temperature; zero is recommended for reproducibility.

top_p Gemini nucleus sampling parameter; currently ignored by OpenAI.

timeout Timeout per attempt, in seconds.

api_key Optional in-memory API key.

project_id Optional OpenAI project id.

max_retries Maximum transient-failure retries.

retry_wait_seconds Initial retry delay.

retry_backoff Retry-delay multiplier.

rate_limit_seconds Minimum delay between requests in the current R process.

4.6 Value

A provider response object containing text, request metadata, model and parameter provenance, hashes, input size, and error details when relevant.

5 Article and dataset workflows

run_article(...)

5.1 Description

Runs one local article through a registered scale prompt. It writes raw audit output plus convenient score and evidence CSV files.

5.2 Important arguments

article_path Local article file.

scale Scale identifier.

provider Provider identifier.

model Model identifier.

output_dir Output directory.

reference_scores Named reviewed scores; activates reference validation.

validation_mode free or reference.

temperature, top_p, timeout Provider parameters.

max_retries, retry_wait_seconds, retry_backoff, rate_limit_seconds Reliability controls.

5.3 Value

A result containing parsed scores, totals, evidence, reasons, validation comparisons, provenance, and output paths.

run_dataset(...)

5.4 Description

Processes all supported local articles in a directory and consolidates scores, evidence, audit information, and errors. A semicolon-separated reference CSV with ID and item columns activates dataset reference mode.

5.5 Value

A dataset result with per-article records and consolidated output paths.

6 Examples

The following example is suitable for an interactive RStudio session:

```
library(scaleLLMflow)
results <- run_dataset(
  articles_dir = "examples/03_pedro_free/articles",
  scale = "pedro",
  provider = "gemini",
  model = "gemini-3.6-flash-lite",
  output_dir = "results/pedro-gemini",
  validation_mode = "free",
  temperature = 0,
  top_p = 0.1
)
```

For comparison against reviewed scores:

```
results <- run_dataset(
  articles_dir = "articles",
  scale = "mqs",
  provider = "openai",
  model = "gpt-4.1-mini",
  output_dir = "results/reference",
  reference_csv = "mqs_reference.csv",
  validation_mode = "reference"
)
```

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